Mohsin Ali

Software Engineer

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Technical Skills

- **Languages**: Python, C#, JavaScript, TypeScript
- **Frameworks**: ASP.NET Core, FastAPI, React, Next.js, Node.js, Azure Functions, Express, Flask, Tailwind CSS
- **Databases**: SQL Server, MySQL, MongoDB, Redis
- **Cloud & DevOps**: Microsoft Azure, Azure Durable Functions, Docker, Kubernetes, Helm, GitHub Actions, Bitbucket Pipelines, Prometheus, Git, Linux
- **AI & LLMs**: Claude and OpenAI APIs, retrieval (RAG) evaluation, embeddings, BM25, prompt and context-window tooling

Professional Experience

SoftBuilds

Software Engineer August 2025 – Present

- Architected and delivered the ASP.NET Core AtHome platform with ShopPro and LabCorp routing integrations, enabling end-to-end order processing and kit fulfillment at scale.
- Enforced HIPAA-aligned data security by implementing Fernet encryption at rest and building secure PHI handling layers across the order processing pipeline.
- Engineered a blob journaling and event-replay system on Azure Storage, enabling full auditability, operation traceability, and point-in-time recovery for critical order workflows.
- Built a reusable .NET utility microservice consolidating Azure Blob Storage access, SAS URL generation, PDF merging, compression, and document transformation into a single shared service layer.
- Developed automated document-generation pipelines for HTML-to-PDF and image-to-PDF conversion, streamlining requisition and fulfillment documentation across healthcare workflows.
- Owned Azure microservices powering LabCorp order fulfillment across **Ditek** (B2B site-based testing) and **Medzah** (direct-to-consumer AtHome testing), supporting distinct operational models on shared infrastructure.
- Built Python Azure Durable Functions orchestrating the full order intake pipeline — from CSV ingestion and requisition generation to shipping label creation, return processing, and refund handling.
- Integrated Endicia and FedEx shipping APIs to fully automate label generation, shipment tracking, and logistics handoffs within the fulfillment pipeline.
- Designed a TRF PDF overlay engine that dynamically stamped order data onto requisition templates, eliminating manual document preparation and accelerating kit dispatch.
- Implemented Node.js Durable Function orchestrators with scheduled triggers to drive multi-stage asset generation pipelines reliably and at scale.
- Owned CI/CD configuration via Bitbucket Pipelines, managing multi-environment deployments across Development, UAT, and Production to ensure consistent, automated releases.

Flippin Technologies

Web Developer January 2025 – July 2025

- Built a full-stack MERN e-commerce platform with end-to-end purchase flows, product management, and Braintree payment integration.
- Designed and implemented a responsive admin dashboard in Next.js and TypeScript, leveraging Tailwind CSS, ShadCN/UI components, and interactive data visualizations.
- Architected a full-stack gamified learning platform delivering progressive educational challenges through interactive, activity-driven modules.
- Secured the platform with JWT authentication and role-based access control, persisting user progress and session state across MongoDB and local storage.
- Developed a suite of educational mini-games — a Sketchbook, a Spelling Bee powered by the Speech Synthesis API, a Shape Matching game via React DnD and Canvas Confetti, and a Reflex Game built on Kaplay.

Open Source

- **StackExchange.Redis** **(contributor, merged)** *(C#)*: fixed batched commands missing from the client's profiler, a bug open since 2022, by sharing one profiling step between single and batched commands; regression test included; merged the same day and shipped in release 3.4.29.
- **graphql-code-generator** **(contributor, merged)** *(TypeScript)*: fixed generated code that did not compile in string mode with no documents, a bug open since 2024; checked the full output with `tsc --strict` on five configurations; shipped in `client-preset` 7.0.0.
- **Authlib** **(contributor, merged)** *(Python, OAuth)*: fixed expired `client_credentials` tokens not being renewed, a bug open since 2024, in the client shared by the Requests and HTTPX integrations; a regression test for each.
- **redis-py** **(contributor, merged)**: documented connection pool limits and `BlockingConnectionPool` after redis-py 8.0 lowered the default to 100 connections; two review rounds with a maintainer.
- **llm-router** *(Python, FastAPI, Redis)*: OpenAI-compatible LLM router with policy routing, fallback, and a daily spending cap that holds under concurrent load through atomic Redis reservations. 236 tests.
- **helmdrift** *(Python, Kubernetes, Helm)*: read-only CLI that finds hand-made changes in a cluster against its Helm chart and names the tool behind each, from `managedFields`. CI runs it on a real kind cluster.
- **semcache** *(Python, Redis)*: semantic cache for LLM calls, in memory or on Redis 8 vector sets, with a benchmark of 250 labelled questions that shows where it starts giving wrong answers. 110 tests; CI on a real Redis and the real embedding model.
- **ragsweep** *(Python)*: sweeps chunking, embeddings and BM25 against a labelled question set to find the retrieval settings that work; 54 runs in 1m 41s, 1.3s when re-run.
- **evalboard** *(TypeScript, Next.js, Recharts)*: in-browser comparison of two eval runs that puts regressions first, where an average hides them; reads ragsweep's results file, and a browser test proves loaded files are never sent anywhere.
- **contextlab** *(TypeScript, Next.js)*: in-browser planner for an LLM context window: what fits, what it costs, and where a long request gets cut.
- **promptlint** *(Python)*: linter for LLM prompt templates that catches broken formatting, missing variables and leaked keys, with autofix.

Education

UET Lahore Lahore

Bachelor in Computer Science